

MATHEMATICS WTW158 / WISKUNDE WTW158
SEMESTER TEST 2 / SEMESTERTOETS 2

Date / Datum: May/Mei 2017 Marks / Punte : 55 Time / Tyd : $2\frac{1}{2}$ hours / ure

MARKS FOR SECTIONS B & C PUNTE VIR AFDELINGS B & C	
MEMO	37

SURNAME / VAN

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

FIRST NAMES / VOORNAME

STUDENT NUMBER

STUDENTENOMMER

--	--	--	--	--	--	--	--

SIGNATURE

HANDTEKENING

CIRCLE YOUR LECTURE GROUP NUMBER / OMKRING JOU LESINGGROEPNOMMER

1	2	3	4	5
Mev Dreyer / Dr Ohlhoff	Prof Harding	Dr Garba	Dr van der Hoff	Dr Jooste

READ THE FOLLOWING INSTRUCTIONS

1. The paper consists of pages 1 to 12. Check whether your paper is complete.
2. **No calculator may be used.**
3. No question paper may be removed from the venue.
4. Any queries about the marking must be done within three days after the tests have been handed back.

LEES DIE VOLGENDE INSTRUKSIES

1. Die vraestel bestaan uit bladsye 1 tot 12. Kontroleer of u vraestel volledig is.
2. **Geen sakrekenaar mag gebruik word nie.**
3. Geen vraestel mag uit die lokaal geneem word nie.
4. Enige navrae oor die nasienwerk moet binne drie dae nadat die toetse teruggegee is, gedoen word.

SECTION A: MULTIPLE CHOICE QUESTIONS - (18 Marks)

1. The questions in Section A must be completed on SIDE 2 of the optical reader form in SOFT PENCIL.
2. First circle your answers on this paper and then transfer these to the optical reader form.

AFDELING A : MEERVOUDIGE KEUSEVRAE - (18 Punte)

1. Die vrae in Afdeling A moet op KANT 2 van die merkleesvorm in SAGTE POTLOOD voltooi word.
2. Omkring eers jou antwoorde op hierdie vraestel en dra dit dan oor na die merkleesvorm.

Question 1 / Vraag 1

The given limit represents the derivative of some function f at some number a .

Die gegewe limiet stel die afgeleide van 'n funksie f voor by 'n sekere getal a .

$$\lim_{h \rightarrow 0} \frac{(1+h)^8 - 1}{h}$$

The function f and the number a is given by: / Die funksie f en die getal a word gegee deur:

(a)	$f(x) = x^9, a = 0$	(b)	$f(x) = x^8 - x, a = 0$	☒ (c)	$f(x) = x^8, a = 1$
(d)	$f(x) = x^8 - 1, a = 1$	(e)	$f(x) = x^7, a = 1$	(f)	None of these / Geen van hierdie

Question 2 / Vraag 2

The slope of the tangent line to the graph of $f(x) = \sin 2x$ at the point $x = \frac{\pi}{2}$ is

Die helling van die raaklyn aan die grafiek van $f(x) = \sin 2x$ in die punt waar $x = \frac{\pi}{2}$, is

(a)	$-\frac{1}{\sqrt{2}}$	☒ (b)	-2	(c)	2
(d)	0	(e)	$-\frac{\sqrt{3}}{2}$	(f)	None of these / Geen van hierdie

Question 3 / Vraag 3

Given / Gegee: $f(x) = \frac{x^2 - 4}{x^2 - 5x + 6}$

The equation(s) of all vertical asymptotes is/are / Die vergelyking(s) van alle vertikale assimptote is

(a)	$x = 3, x = 2$	(b)	$y = 3$	(c)	$y = 2, y = 3$
☒ (d)	$x = 3$	(e)	$x = 2$	(f)	None of these / Geen van hierdie

Question 4 / Vraag 4

Determine / Bepaal $\lim_{\theta \rightarrow 0} \frac{2 \tan \theta}{2\theta + \sin 3\theta}$

(a)	0	(b)	$\frac{2}{5\theta}$	(c)	$\frac{2}{3}$
(d)	$\frac{5\theta}{2}$	(e)	$\frac{2}{5}$	(f)	None of these / Geen van hierdie

Question 5 / Vraag 5

If / As $f(x) = \sinh(x^2 - 3x)$ then / dan is $f'(x) =$

(a)	$(2x - 3) \cosh(x^2 - 3x)$	(b)	$(3 - 2x) \sinh(x^2 - 3x)$	(c)	$\sinh(x^2 - 3x)$
(d)	$(3 - 2x) \cosh(x^2 - 3x)$	(e)	$-\sinh(2x - 3)$	(f)	None of these / Geen van hierdie

Question 6 / Vraag 6

Suppose that $f(2) = 3$, $f'(2) = 4$, $g(2) = -1$ and $g'(2) = 2$. Then $(fg)'(2)$ is equal to

Veronderstel dat $f(2) = 3$, $f'(2) = 4$, $g(2) = -1$ en $g'(2) = 2$. Dan is $(fg)'(2)$ gelyk aan

(a)	2	(b)	-2	(c)	8
(d)	3	(e)	$\frac{9}{2}$	(f)	None of these / Geen van hierdie

Question 7 / Vraag 7

If / As $f(x) = x \ln x - x$ then / dan is $f''(x) =$

(a)	$\ln(x - 1)$	(b)	$-\frac{1}{x^2}$	(c)	$-\frac{1}{x}$
(d)	$\ln x - 1$	(e)	$\frac{1}{x}$	(f)	None of these / Geen van hierdie

Question 8 / Vraag 8

The point on the curve $y = \frac{e^x}{x+2}$ at which the tangent line is horizontal is

Die punt op die kromme $y = \frac{e^x}{x+2}$ waar die raaklyn horisontaal is, is

(a)	(2, 0)	(b)	$(1, \frac{e}{2})$	(c)	$(-1, \frac{1}{e})$
(d)	(0, 1)	(e)	None of these / Geen van hierdie		

Question 9 / Vraag 9

The value(s) of c that satisfy the Mean Value Theorem for the function $f(x) = x - x^3$ on the interval $[0, 2]$ is given by

Die waarde(s) van c wat die Middelwaardestelling bevredig vir die funksie $f(x) = x - x^3$ op die interval $[0, 2]$ word gegee deur

(a)	$\frac{2}{\sqrt{3}}$ and / en $-\frac{2}{\sqrt{3}}$	(b)	$\frac{\sqrt{3}}{2}$ and / en $-\frac{\sqrt{3}}{2}$	(c)	$\frac{4}{3}$
(d)	$\frac{2}{\sqrt{3}}$	(e)	None of these / Geen van hierdie		

Question 10 / Vraag 10

Which of the following statement(s) is/are always true? / Watter van die volgende stelling(s) is altyd waar?

- I If $f'(c) = 0$, then $f(x)$ has a maximum or minimum value at $x = c$.
As $f'(c) = 0$, dan het $f(x)$ 'n maksimum of minimumwaarde by $x = c$.
- II If $f'(x) = g'(x)$ for all x in the interval I , then $f(x) = g(x)$ on I .
As $f'(x) = g'(x)$ vir alle x in die interval I , dan is $f(x) = g(x)$ op I .
- III If $f(x)$ is differentiable on the open interval (a, b) , and c is a local maximum of f in (a, b) , then $f'(c) = 0$.
As $f(x)$ differensieerbaar is op die oop interval (a, b) , en c is 'n lokale maksimum vir f op (a, b) , dan is $f'(c) = 0$.

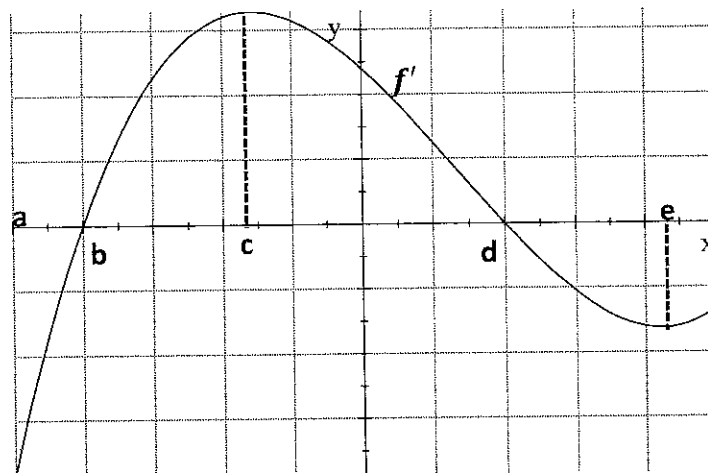
(a)	I only / Slegs I	(b)	II only / Slegs II
(c)	III only / Slegs III	(d)	I and II only / Slegs I en II
(e)	I and III only / Slegs I en III	(f)	I, II and III / I, II en III

Questions 11 and 12 refer to the following graph:

Vrae 11 en 12 verwys na die volgende grafiek:

The graph of the first derivative f' of a function f is given.

Die grafiek van die eerste afgeleide f' van 'n funksie f is gegee.



Question 11 / Vraag 11

The function f has a local minimum at: / Die funksie f het 'n lokale minimum by:

(a)	a	☒ (b)	b	(c)	d
(d)	e	(e)	None of these / Geen van hierdie		

Question 12 / Vraag 12

The function f is concave down on the interval: / Die funksie f is konkaf na onder op die interval:

(a)	(a, c)	(b)	(b, c)	(c)	(b, d)
☒ (d)	(c, e)	(e)	None of these / Geen van hierdie		

18

SECTION B (15 Marks) / AFDELING B (15 Punte):

- The questions in Section B must be completed on this paper in pen.
- Only the answer in the provided box will be marked.
- Calculations can be performed on the facing empty pages and **will not be marked**.

- Die vrae in Afdeling B moet op die vraestel voltooi word in pen.
- Slegs die antwoord in die gegewe blok sal gemerk word.
- Berekeninge kan op die oop teenblaaië gedoen word en **sal nie gemerk word nie**.

Question 1 / Vraag 1

Let f be the function defined by / Laat f die funksie wees gedefinieer deur

$$f(x) = \begin{cases} kx^2 - 1 & \text{if / as } x \geq 2 \\ k^2x - 1 & \text{if / as } x < 2 \end{cases}$$

Give the value(s) of k such that the function f is continuous at $x = 2$.

Gee die waarde(s) van k sodat die funksie f kontinu is by $x = 2$.

$$k = 0 \quad \text{or} \quad k = 2$$

2

Question 2 / Vraag 2

$$\lim_{x \rightarrow \infty} \frac{9 - |x|}{9 - x} = \dots \text{ (if it exists. / indien dit bestaan.)}$$

$$1$$

1

Question 3 / Vraag 3

For which value(s) of x is the function $f(x) = |x^2 - 4|$ not differentiable?

Vir watter waarde(s) van x is die funksie $f(x) = |x^2 - 4|$ nie differensieerbaar nie?

$$x = -2, \quad x = 2$$

2

Question 4 / Vraag 4

Find the derivatives of the following. Do not simplify your answers.

Bepaal die afgeleides van die volgende. Moenie jou antwoord vereenvoudig nie.

4.1 $f(x) = x^2 e^{\tan x}$

$$f'(x) = 2x e^{\tan x} + x^2 e^{\tan x} \cdot \sec^2 x$$

4.2 $f(x) = \arctan(x^2 + 1) = \text{bgtan}(x^2 + 1) = \tan^{-1}(x^2 + 1)$

$$f'(x) = \frac{2x}{1 + (x^2 + 1)^2}$$

Question 5 / Vraag 5

Given that / Gegee dat $f(\theta) = \theta + \sin \theta$ on / op $[0, 2\pi]$.

For what value(s) of θ will f be concave down? / Vir watter waarde(s) van θ is f konkaf na onder?

$$\theta \in (0, \pi)$$

Question 6 / Vraag 6

Evaluate / Bepaal:

$$\lim_{x \rightarrow 0} \frac{x - \sin x}{x^2} =$$

$$0$$

Question 7 / Vraag 7

Let f be a function with $f'(x) = 1 - \frac{1}{1-x}$. On which interval(s) is f increasing?

Laat f 'n funksie wees met $f'(x) = 1 - \frac{1}{1-x}$. Op watter interval(le) sal f stygend wees?

$$x \in (-\infty, 0) \cup (1, \infty)$$

SECTION C (22 Marks) / AFDELING C (22 Punte):

1. The question in Section C must be completed on this paper.
2. All calculations must be shown and will be marked.
3. If you need more space for an answer, use the facing page and indicate it clearly by framing it.
4. Attention should be paid to writing in a mathematically sound way.

1. Die vrae in Afdeling C moet op die vraestel voltooi word.
2. Alle berekeninge moet getoon word en sal gemerk word.
3. As jy nog ruimte vir 'n antwoord nodig het, gebruik die teenblad en dui dit duidelik aan deur 'n raam daarom te trek.
4. Maak seker dat jou skryfwyse wiskundig korrek is.

Question 1 / Vraag 1

1.1 If $y = (\ln x)^x$ determine $\frac{dy}{dx}$. / As $y = (\ln x)^x$ bepaal $\frac{dy}{dx}$.

$$\begin{aligned}\ln y &= x \ln(\ln x) \\ \Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} &= \ln(\ln x) + x \cdot \frac{\frac{1}{x}}{\ln x} \\ \Rightarrow \frac{dy}{dx} &= y \left(\ln(\ln x) + \frac{1}{\ln x} \right) \\ \Rightarrow \frac{dy}{dx} &= (\ln x)^x \left[\ln(\ln x) + \frac{1}{\ln x} \right]\end{aligned}$$

1.2 Determine $\frac{dy}{dx}$ if $e^{xy} = \frac{y^2}{x}$. / Bepaal $\frac{dy}{dx}$ as $e^{xy} = \frac{y^2}{x}$.

$$\begin{aligned}e^{xy} \left(y + x \frac{dy}{dx} \right) &= \frac{x \cdot 2y \cdot \frac{dy}{dx} - y^2}{x^2} \\ \Rightarrow x^2 y e^{xy} + x^3 \cdot e^{xy} \frac{dy}{dx} &= 2xy \cdot \frac{dy}{dx} - y^2 \\ \Rightarrow (x^3 \cdot e^{xy} - 2xy) \frac{dy}{dx} &= -y^2 - x^2 y e^{xy} \\ \Rightarrow \frac{dy}{dx} &= \frac{y^2 + x^2 y \cdot e^{xy}}{2xy - x^3 \cdot e^{xy}}\end{aligned}$$

1.3 Use logarithmic differentiation to determine $\frac{dy}{dx}$ if / Gebruik logaritmiese differensiasie en bepaal $\frac{dy}{dx}$ as

$$y = \frac{\sqrt{\sec(x+1)}}{6^x (x^2-1)^4}$$

$$\ln y = \frac{1}{2} \ln \sec(x+1) - \ln 6^x - 4 \ln(x^2-1)$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \frac{1}{2} \frac{\sec(x+1) \cdot \tan(x+1)}{\sec(x+1)} - \frac{6^x \cdot \ln 6}{6^x} - 4 \cdot \frac{2x}{x^2-1}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sqrt{\sec(x+1)}}{6^x \cdot (x^2-1)^4} \left[\frac{1}{2} \tan(x+1) - \ln 6 - \frac{8x}{x^2-1} \right]$$

3

Question 2 / Vraag 2

Evaluate the given limit, if it exists. / Bepaal die gegewe limiet, indien dit bestaan.

$$\lim_{x \rightarrow -\infty} x^2 e^x$$

$$\begin{aligned} & \lim_{x \rightarrow -\infty} x^2 \cdot e^x \\ &= \lim_{x \rightarrow -\infty} \frac{x^2}{e^{-x}} \quad \left(\frac{\infty}{\infty} \right) \\ &= \lim_{x \rightarrow -\infty} \frac{2x}{-e^{-x}} \quad \left(\frac{-\infty}{-\infty} \right) \\ &= \lim_{x \rightarrow -\infty} \frac{2}{e^{-x}} \quad \left(\frac{c}{\infty} \right) \\ &= 0 \end{aligned}$$

2

Question 3 / Vraag 3

Use the Closed Interval Method to find the extreme values of $f(x) = \frac{1}{x^2 + x + 1}$ on the interval $[-2, 1]$.

Gebruik die Geslote Intervalmethode en vind die ekstreemwaardes van $f(x) = \frac{1}{x^2 + x + 1}$ op die interval $[-2, 1]$.

f is continuous on $[-2, 1]$

Critical numbers:

$$f'(x) = \frac{-(2x+1)}{(x^2+x+1)^2} \Rightarrow f'(x) = 0 \text{ if } x = -\frac{1}{2}$$



$$f(-2) = \frac{1}{3}$$

$$f(-\frac{1}{2}) = \frac{4}{3}$$

$$f(1) = \frac{1}{3}$$

f has a global min of $\frac{1}{3}$ at $x = -2$ and at $x = 1$

f has a global max of $\frac{4}{3}$ at $x = -\frac{1}{2}$.

3

Question 4 / Vraag 4

Given that / Gegee dat $f(x) = xe^{-x}$.

Use the first derivative test to determine the local extreme value(s) of the function.

Gebruik die eerste-afgeleide toets en bepaal die lokale ekstreemwaarde(s) van die funksie.

$$\begin{aligned} f'(x) &= e^{-x} + x \cdot e^{-x}(-1) \\ &= (1-x)e^{-x} \end{aligned}$$

$$f'(x) = 0 \Rightarrow x = 1$$



f has a local maximum $(1, e)$.

3

Question 5 / Vraag 5

5.1 Complete the following definition: / Voltooi die volgende definisie:

The number c is a critical number of f if.....

Die getal c is 'n kritieke getal van f indien.....

$c \in Df$ such that
 $f'(c) = 0$ or $f'(c)$ does not exist

5.2 Determine the critical numbers of / Bepaal die kritieke getalle van

$$f(x) = \frac{x^2}{x-1}$$

$$\begin{aligned} f'(x) &= \frac{2x(x-1) - x^2}{(x-1)^2} \\ &= \frac{x^2 - 2x}{(x-1)^2} \\ &= \frac{x(x-2)}{(x-1)^2} \end{aligned}$$

$$f'(x) = 0 \Rightarrow x = 0 \text{ or } x = 2$$

$f'(x)$ does not exist when $x = 1$
but $1 \notin Df$

So the critical numbers are
 $c = 0$, $c = 2$.

Question 6 / Vraag 6

Prove that if f is differentiable at $x = a$ then f is continuous at $x = a$.

Bewys dat as f differensieerbaar is by $x = a$ dan is f kontinu by $x = a$.

We know that f is differentiable in a
 $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ is finite

We need to prove that f is continuous
in $x = a$, that is $\lim_{x \rightarrow a} f(x) = f(a)$

$$\lim_{x \rightarrow a} f(x)$$

$$= \lim_{x \rightarrow a} (f(x) - f(a) + f(a))$$

$$= \lim_{x \rightarrow a} (f(x) - f(a)) + \lim_{x \rightarrow a} f(a)$$

$$= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot (x - a) + \lim_{x \rightarrow a} f(a)$$

$$= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot \lim_{x \rightarrow a} (x - a) + f(a)$$

$$= f'(a) \cdot 0 + f(a)$$

$$= 0 + f(a)$$

$$= f(a)$$